The Cost of Early Greening: Structural Overshoot Amplifies Seasonal Water Deficit in Northeastern Siberian Tundra

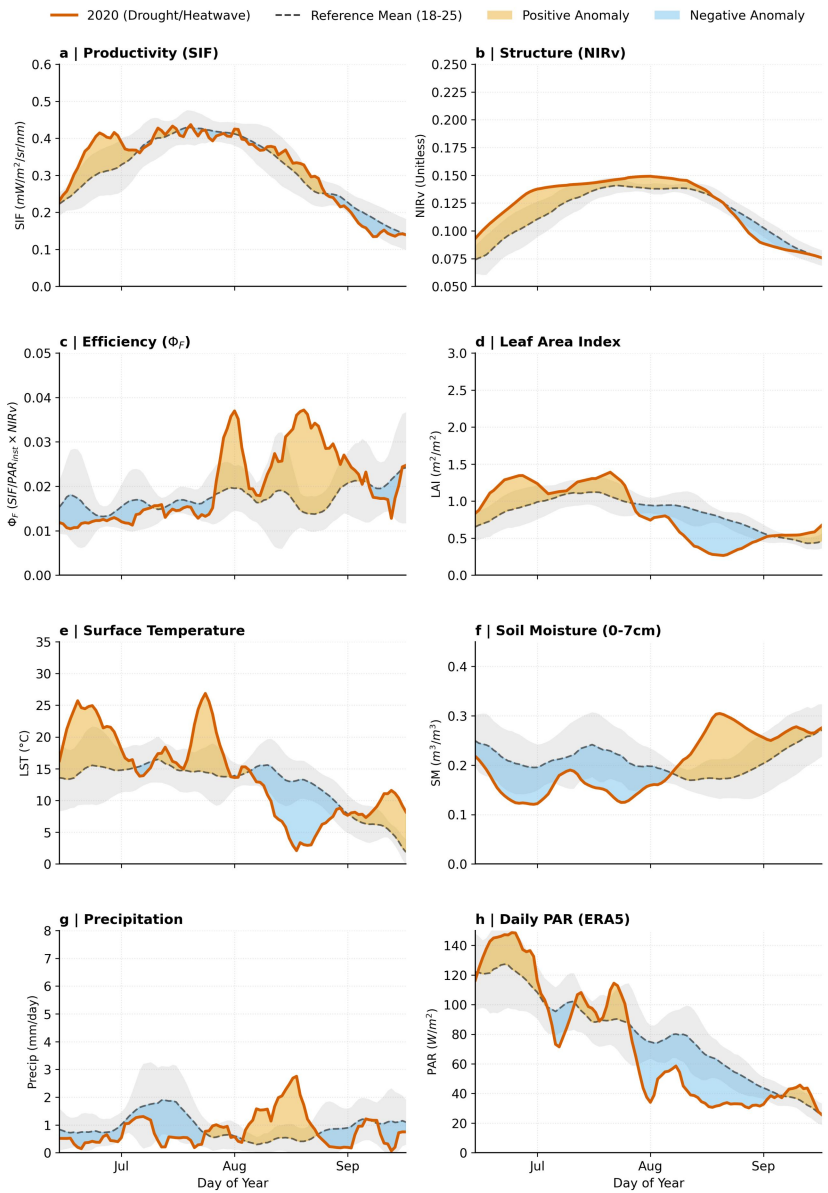

Figure Caption:

The study area covers the Indigirka Lowlands (140.0°–150.0°E, 70.0°–72.0°N), excluding the complex mountainous terrain in the southeast. Time series display the daily evolution (7-day rolling mean) of key ecosystem variables for the 2020 growing season (solid colored lines) compared to the multi-year reference mean (2018–2025, dashed grey lines). Shaded areas indicate positive (orange) and negative (blue) anomalies. (a–d) Vegetation Dynamics: The panels reveal a structural-physiological decoupling: while structural proxies (b) NIRv and (d) LAI showed positive anomalies (structural overshoot) during the early warming phase, physiological efficiency (c) Phi_F (calculated as SIF / (PAR_inst * NIRv)) (e–h) Environmental Drivers: The heatwave dynamics are characterized by: (e) LST anomalies confirming the extreme heat, which, combined with low (g) precipitation, led to a severe deficit in (f) top-layer soil moisture (0–7 cm).

Preliminary Results (Draft):

Preliminary analysis reveals a mechanism of "Structural Overshoot" driving the ecosystem collapse.

We identified three distinct phases in the vegetation response to the 2020 heatwave. First, during the early warming (June–July), the ecosystem experienced a "false boom" or Structural Overshoot: the high temperatures triggered rapid canopy expansion (positive anomalies in NIRv (b) and LAI (d)). However, this excessive growth accelerated soil moisture depletion, creating a "hydraulic debt" while physiological efficiency (Φ_F) was already being suppressed.

Second, a critical Decoupling/Crash occurred in late July (DOY 198-210). As soil moisture hit its minimum, SIF collapsed sharply due to hydraulic failure, while NIRv (structure) lagged behind, still showing high values before eventually diving.

Finally, we observed an Irreversible Scar in August. Although environmental conditions improved (soil moisture recovered) and physiological function (Φ_F) bounced back to positive anomalies, the canopy structure (LAI) could not recover. The ecosystem had become "smaller and sparser" due to the earlier overshoot, losing the potential for high productivity in late summer.